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Machine learning with a reduced dimensionality representation of comprehensive Pentacam tomography parameters to identify subclinical keratoconus

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ABSTRACT

Purpose: To investigate the performance of a machine learning model based on a reduced dimensionality parameter space derived from complete Pentacam parameters to identify subclinical keratoconus (KC).
Methods: All 1692 available parameters were obtained from the Pentacam imaging machine on 145 subclinical KC and 122 control eyes. We applied a principal component analysis (PCA) to the complete Pentacam dataset to reduce its parameter dimensionality. Subsequently, we investigated machine learning performance of the random forest algorithm with increasing numbers of components to identify their optimal number for detecting subclinical KC from control eyes.
Results: The dimensionality of the complete set of 1692 Pentacam parameters was reduced to 267 principal components using PCA. Subsequent selection of 15 of these principal components explained over 85% of the variance of the original Pentacam-derived parameters and input to train a random forest machine learning model to achieve the best accuracy of 98% in detecting subclinical KC eyes. The model established also reached a high sensitivity of 97% in identification of subclinical KC and a specificity of 98% in recognizing control eyes.
Conclusions: A random forest-based model trained using a modest number of components derived from a reduced dimensionality representation of complete Pentacam system parameters allowed for high accuracy of subclinical KC identification.

1. Introduction

Keratoconus (KC) is a progressive, bilateral and asymmetric corneal condition characterised by abnormal thinning and protrusion of the cornea, with resulting significant impairment in vision [1]. Although damage from KC is irreversible, early treatment including the use of corneal collagen cross-linking (CXL), may slow its progression and therefore limit subsequent impact on visual impairment [2,3]. Detection of subclinical KC is essential to ensure that KC patients are identified as early as possible to avoid the need for other optical interventions or their potential progression for a corneal transplant [4].

Corneal tomography systems such as Pentacam, GALILEI and Sirius are used in the diagnosis of KC. These instruments allow trained clinicians to not only identify advanced forms of KC but are also used in providing data for the detection of subclinical KC. Subclinical KC, on the other hand, is usually defined as an eye that exhibits certain KC topographic features but is not severe enough to be classified as clinically manifested KC [5]. Its fellow eye may be either a clinically manifested KC or a normal eye. The diagnosis is achieved through assessment of corneal topographic maps produced from these machines. While human interpretation of the complex patterns obtained through these corneal topographic maps is readily achievable for the more advanced forms of

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KC, there is an element of subjectivity at the subclinical level leading to low reproducibility.

In an attempt to address subjective variability, a number of automated machine learning models have been published with the aim of providing a more objective approach to the identification of subclinical KC from control eyes [6–18]. The majority of these studies have used parameters derived from a particular topographic or tomographic imaging system and subsequently built machine learning models for the diagnosis of subclinical KC [6–14,18]. However, these models have typically been based on selection of a subset of parameters that were manually decided by the authors based on prior clinical features. Concomitant with this is the observation that the parameters used to produce these models vary between studies and are also subjective [6–14,18].

As an example, the Pentacam corneal tomography system, which is a commonly used imaging machine in detecting KC, has a total of 1692 parameter measurements potentially available. A recent study by Ruiz et al. [11], selected only 22 Pentacam parameters for incorporation into their support vector machine model. This model achieved an accuracy of 93%, a sensitivity of 79% and a specificity of 98% in detecting subclinical KC over their data set. The majority of the parameters chosen in their study had previously been commonly considered in KC, e.g. radii of curvature in the horizontal and vertical meridians (Rh, Rv), minimum corneal thickness and anterior chamber depth (ACD). While the reported performance of the model was high, manual selection of specific parameters is a discretionary decision, and it remains unclear whether incorporation of other parameters would aid in the detection of subclinical KC. We suggest that it is important to more systematically consider the incorporation of previously unexplored parameters into models to investigate their effectiveness in the identification of subclinical KC. A reduction from 1692 to 22 parameters omits consideration of the majority of available parameters and thus there may be substantial information loss in detecting subclinical KC.

Few machine learning models have been applied to analysis of Pentacam data, but these have typically relied on the use of a small subset of chosen parameters [10,13,14]. To the best of our knowledge, no studies have evaluated the effectiveness of a complete set of Pentacam parameters in the detection of subclinical KC. In part, this may reflect the complexity of the data and potential redundancy between different parameters that may introduce noise into any subsequent analysis. To address this issue, we undertook a reduction in the dimensionality of the data through the use of a principal component analysis (PCA) approach. This allowed noise of the data, e.g. highly correlated parameters, to be identified before remaining parameters were presented to a subsequent machine learning method.

The aim of this study was therefore to investigate the performance of a machine learning model where the complete output of these derived parameters could be used in a machine learning space to detect subclinical KC.

2. Materials and methods

This study was approved by the Royal Victorian Eye and Ear Hospital (RVEEH) Human Research and Ethics Committee (Project #10/954H). The protocol followed the tenets of the Declaration of Helsinki and all privacy requirements were met. Details of this study regarding patient recruitment and study methods have been previously published [19–22].

2.1. Subjects

In the present study, we downloaded all Pentacam parameters from participants without prior pre-selection of parameters. The Pentacam software version 1.20r127 (Oculus, Wetzlar, Germany) was used for this purpose. The data collection was conducted at the RVEEH from participants between 2007 and 2019.

Subclinical KC is sometimes referred to as *forme fruste* KC (FFKC), while there is no generally recognized definition of subclinical KC or FFKC in the literature or in clinics [5] (In the literature, the FFKC is typically defined as one eye that is normal in a KC patient who simultaneously has clinically apparent KC in the other eye). We adopt the term subclinical KC, which is routinely used by the clinicians at the Royal Victoria Eye and Ear Hospital (RVEEH).

Subclinical KC subjects were defined as those eyes with abnormal corneal tomography, including inferior-superior localized steepening or an asymmetric bowtie pattern, but without detectable clinical signs on slit-lamp biomicroscopy and retinoscopy examination [19–22]. This assessment was undertaken by an experienced optometrist (SS) together with RVEEH KC specialists and only eyes consistently identified as subclinical KC by all examiners were included as cases in this study. Eyes with other corneal abnormalities such as corneal degenerations and keratitis were excluded from the recruited group.

The control group consisted of eyes that had no known history of any corneal disorder, but may have presented with another ocular condition, such as cataract, myopia, retinal disease, macular scar, macular oedema or retinal dystrophy and amblyopia that did not introduce a corneal change. The diagnosis and medical history of subjects were obtained from patients' medical records. Eyes with other ocular conditions that may affect the anterior segment of the eye were excluded from the control group.

After selection and labelling, the dataset consisted of 267 eyes in total of which 145 (54.3%) had subclinical KC from 141 participants (97% were unilateral subclinical KC) whereas 122 eyes (45.7%) were control eyes (no KC) from a total of 85 participants. For each participant, either one eye or both eyes, that satisfied the inclusion criteria, were included in the study.

2.2. Analysis steps

The analysis can be divided into the following steps: data cleaning, dimensionality reduction, training, testing, and evaluation. The details of these steps are presented in the flowchart (Fig. 1). All analyses were configured using RStudio Server Pro (Version 1.3.1056-1) for Windows, including data cleaning, dimensionality reduction, and random forest machine learning procedures. Dimensionality reduction was accomplished using the prcomp function of the stat package (version 4.0.3). The caret package [23] (version 6.0.88) was used to conduct all machine learning processes, including training and assessing the performance

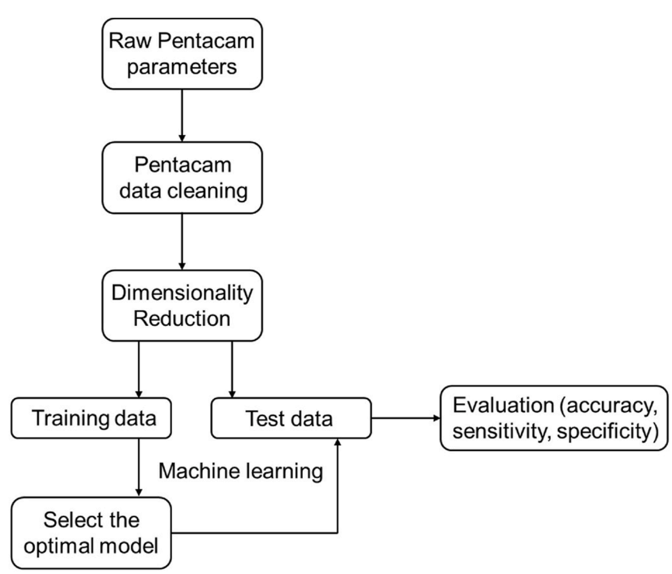

Fig. 1. Flow diagram of steps taken for the analysis of subclinical KC and control eyes.

(testing) of created models. The 'train' function of the caret package automatically creates models for each set of principal components, presents performance through 10-fold cross validation, and fine-tunes the hyper-parameters in each random forest model under consideration. The tuning parameters were chosen based on the model's accuracy, and the 'predict' function was then used to assess the final model using the test dataset. The code for dimensionality reduction, model establishment, tuning, and evaluation has been published on GitHub and is available at https://github.com/doris-coder/Random-forest-model-using-reduced-dimensionality-parameter-space-derived-from-Pentacam-parameters.git.

2.3. Data cleaning

In this process, low quality, irrelevant/non-generalisable, duplicated, incomplete, uninformative or imprecise parameters were identified and then either replaced or removed from the dataset (Supplementary Figure 1). The following sections detail each step of data cleaning:

- Low-quality parameters (n = 475): Pentacam output files include information on each parameter's image quality, which may be utilized to detect low-quality parameters. "In all eyes examined, only parameters with the quality label "OK" were analysed; those with other labels including "Blinking Error", "Calc Error Scheimpflug! ", "Lid closure! ", and "Blinking Error / Nose Shadow!" were excluded.".
- Irrelevant/non-generalisable (n = 75): Irrelevant parameters, including patient identifiers such as first and last names, as well as examination identifiers such as examination time, date, comments, and image quality, are removed from the analysis.
- Duplicated parameters (n = 165): Multiple parameters that contained the same value were identified as duplicated parameters. One parameter was retained for use in the analysis for each duplicated parameter set found.
- Incomplete parameters: If a parameter was completely blank or had a missing rate of more than 20%, it was excluded from the analysis (n = 138). Missing values were replaced with the mean of the non-missing values for parameters having a missing rate of less than 20%.
- Uninformative parameters (n = 29): Parameters that had the same value or text throughout the whole dataset were excluded from the analysis since they offered no useful information.
- Imprecise parameters: Minimum and maximum values for each parameter were determined using the Pentacam interpretation guidebook, the Pentacam user's manual, and a literature search.

2.4. Dimensionality reduction

Dimensionality reduction was conducted for the given set of cleaned Pentacam data. The principal component analysis (PCA) was used for this process. PCA is one of the most fundamental tools for dimensionality reduction, and is effective for extracting key parameters from high dimensional input data [24]. Most importantly, using PCA mapping, the high-dimensional dataset was reduced to a low-dimensional dataset while at the same time maintaining the information integrity of the dataset.

Pentacam parameters derived from the output of the PCA method provided a set of new parameters, known as principal components, where each component represented a linear combination of parameters in the original set of features.

2.5. Training and test data

The dataset was split into a training and a test dataset. This was achieved by randomly selecting 70% eyes (case and control) as a training dataset and the remaining 30% of eyes (case and control) provided the test data. The training dataset was used to build machine learning models and the optimal PCA model via a 10-fold cross validation procedure; while the test data were applied to assess the performance of the final fitted model.

2.6. Selecting the optimal model

Following the PCA data dimensionality reduction, we applied the random forest method to identify subclinical KC eyes. We chose to use the random forest method as our machine learning method as we have previously reported that this method provided superior performance compared to 7 other machine learning methods (i.e. Support vector machine, K-nearest neighbours, Logistic regression, Linear discriminant analysis, Lasso regression, Decision tree and Multilayer perceptron neural network) in the detection of subclinical KC based on a previous study utilizing a smaller number of only 11 Pentacam parameters [25].

To investigate how many principal components were important in detecting subclinical KC, we first ordered the components based on the amount of variance in the data explained by each component. Random forest models were built using the first two principal components and other principal components were added one after the other until there was no improvement in the model. This sequential operation allowed acquisition of the most important component set that had the best performance in detecting subclinical KC.

2.7. Evaluation of the final model

To objectively investigate the final established model from the training model, we undertook analysis in the test dataset and assessed performance based on three commonly employed evaluation measures, i.e. accuracy, sensitivity and specificity.

3. Results

In the initial training dataset using 70% of the available samples, a total of 186 eyes were randomly selected consisting of 107 (57.5%) subclinical KC eyes and 79 (42.5%) control eyes. The test dataset consisted of the remaining eyes being 38 (46.9%) subclinical KC eyes and 43 (53.1%) control eyes.

Following Pentacam data cleaning, the number of available parameters reduced from 1692 to 810 parameters for subsequent use in the machine learning aspect of the study (Supplementary figure 1). According to the clinical interpretation given in the Pentacam user guidebook, parameters included in the analysis fell into 3 main clinical groups: corneal power (490 parameters), corneal thickness (235 parameters) and other (85 parameters). The "other" parameter group consisted of parameters related to other corneal measures such as corneal volume. Supplementary Figure 3 shows the example of each parameter group within the Pentacam output following pre-processing.

PCA further reduced the number of parameters from 810 to 267 principal components in the complete dataset. Fig. 2 shows the proportion of variation explained (i.e. proportion of variance of the original Pentacam-derived parameters) by each of the first 20 principal components. A total of 87.5% of the information (variance) contained in the data were retained by the first 20 principal components. A total of 100% of the information (variance) contained in the data were retained by the first 264 principal components, and each of the remaining 3 components contained less than 0.001% of the information of the data.

To objectively decide how many principal components were optimal for detecting subclinical KC, we first examined the combination of the first two principal components (accounting for 49% of the variance) and then sequentially added more principal components one by one. Fig. 3 shows the accuracy of differentiating subclinical KC from controls using each principal component combination. The accuracy is reported in terms of maximum, mean, and minimum accuracy across a 10-fold cross validation. A combination of the first 15 principal components provided the highest mean discriminative accuracy of 97% (a sensitivity of 96%

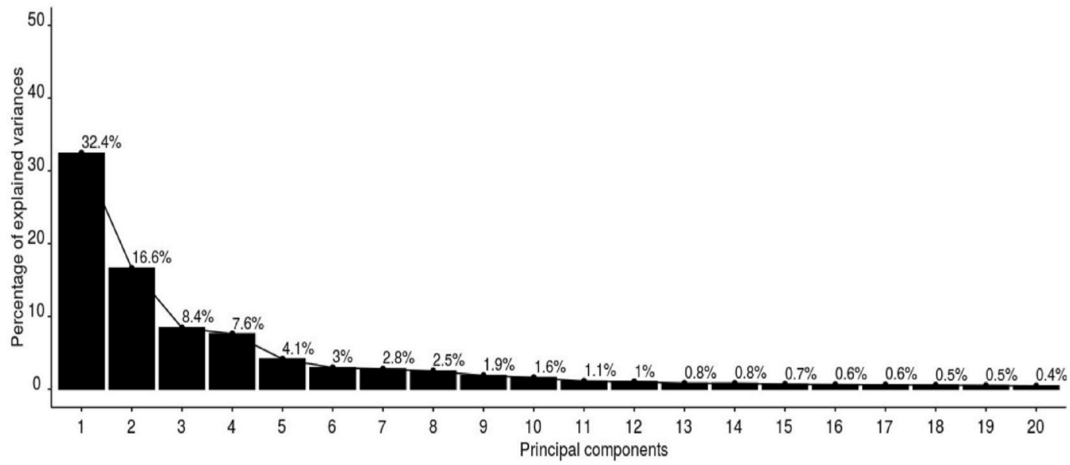

Fig. 2. The proportion of variance explained by each of the first 20 principal components. X-axis indicates each of the 20 principal components.

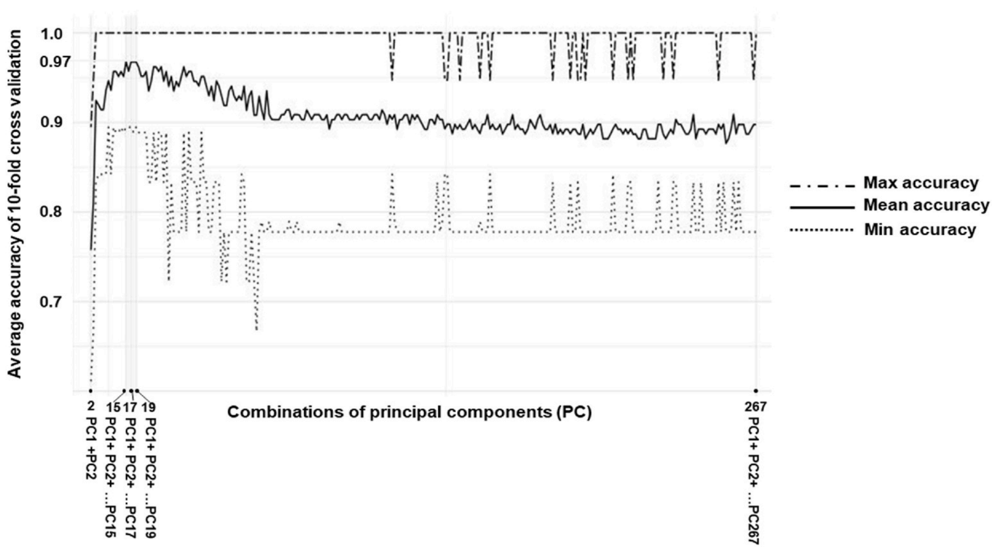

Fig. 3. Accuracy in detecting subclinical KC using different principal component combinations. A combination of the first two principal components were used followed by sequential addition of each of the remaining 265 principal components. X-axis indicates the number of principal components in each combination. Y-axis indicates accuracy of detection of subclinical KC. Accuracy was obtained by calculating maximum, mean, and minimum accuracy across 10-fold cross validation in the training data.

and a specificity of 98%) between subclinical KC and control eyes in the training dataset. When applying 10-fold cross-validation to the first 15 principal components, the maximum accuracy was 100% and the minimum accuracy was 89%. The same performance was also obtained using either the first 17 principal components or the first 19 principal components, but performance became suboptimal when using the first 18 principal components (mean accuracy of 97%, maximum accuracy across folds of 100% and minimum accuracy across folds of 88%). In summary, using the first 15 principal components provided the most parsimonious model for maximal performance in detecting subclinical KC from controls.

To objectively assess the performance of the derived machine learning model on generalization to unseen data, we computed the accuracy, sensitivity and specificity on the remaining 30% of the data that acted as an independent validation dataset. The accuracy in discriminating subclinical KC eyes from control eyes slightly improved to 98% compared to the training dataset, sensitivity of identifying subclinical KC eyes also improved slightly from 96% to 97%, and the specificity of detecting control eyes remained the same as the training data at 98% (Table 1).

Table 1
Accuracy, sensitivity and specificity of the model built in training and test data. The mean ± standard deviation of each measure over the folds is recorded for the training dataset.

	Training dataset	Test dataset
Total No. of eyes	186	81
No. of subclinical eyes	107	38
No. of control eyes	79	43
Accuracy	97% ± 4%	98%
Sensitivity	96% ± 6%	97%
Specificity	98% ± 5%	98%

4. Discussion

In this study, we established a machine learning model that utilized the complete output of parameters from a Pentacam tomographic machine. The model could automatically identify subclinical KC with high accuracy in both a training dataset as well as an independent test dataset. The derived model presented with high accuracy, sensitivity and specificity, demonstrating broad generalizability of our model when presented with a new dataset. This model has potential applicability of being used as a Pentacam based subclinical KC detection index.

Over recent years, numerous studies have applied various machine learning methods to parameters derived from various corneal topography systems in order to detect subclinical KC. A fundamental step in the development of any machine learning model is the training process by which the machine learning algorithm "learns" input parameters to make a correct diagnosis. Essentially, the machine learning algorithm can only use information that has been used to train it, thus utilizing all available parameters from any imaging machine would make the most use of such data.

Previous studies have typically relied on a manual selection or inclusion of only a limited number of parameters (ranging from 22 to 55) [6–8,11,18]) generated from a corneal topography system and would therefore likely limit the applicability of any model. This would occur through potential bias of the selected parameters or through exclusion of other parameters that could enhance the model. For example, corneal curvature derived parameters, such as mean keratometry (MeanK), maximum keratometry (Kmax), Rh and Rv are commonly considered parameters in identifying KC. In the study by Smadja et al. [7], 55 parameters were derived from the GALILEI Dual system and used to build a decision tree model for the identification of subclinical KC from clinically determined control eyes. Their study included both anterior, and posterior corneal curvature related parameters such as, Kmax, Axial MeanK (0–4 mm), Tang MeanK (4–7 mm), Eccentricity, GALILEI derived indices: differential sector index (DSI) and irregular astigmatism index (IAI). In comparison, the study by Ruiz et al. [11] included 22 Pentacam derived parameters to establish a support vector machine model for the detection of subclinical KC. In their study, they selected a different set of curvature related parameters, e.g., Rh, Rv, Kmax, vertical location of Kmax from the corneal apex (KmaxY), eccentricity in superior (Ecc Sup) and inferior (Ecc Inf) meridians. Although there was some overlap in the parameters chosen between the two studies, the manual selection of specific parameters that are deemed to be important represents a subjective choice. It is preferable to develop a model in an unbiased, empirical, manner.

For these reasons, we argue that it is necessary to explore the use of all available parameters to obtain the most accurate detection model for subclinical KC. A caveat to the inclusion of all data is that additional parameters above and beyond those that are strictly necessary for generating the most parsimonious model may introduce noise into the model. The present study sought to develop and use a machine learning model using a complete parameter set generated from a Pentacam system but also limit the impact of potential noise provided by superfluous parameters. This was achieved through the use of a PCA allowing for the removal of redundant or overlapping parameters in order to achieve the highest accuracy in differentiating subclinical KC from control eyes.

Compared to previous machine learning models in detecting subclinical KC from control eyes, the model developed in the present study had the best accuracy, and a comparable, if not, higher sensitivity and specificity compared to prior models (Table 2). Our model also showed consistent performance on different evaluation measures, meaning that it had a high ability to detect both subclinical KC eyes as well as to detect control eyes. Thus, the model built in our study could be employed not only by KC clinics, to support the identification of subclinical KC cases, but also by refractive surgery clinics, to rule out subclinical KC in potential refractive surgery candidates. The high performance achieved in this study indicated that the inclusion of the entire output from a Pentacam improves the differentiable power of both subclinical KC and control eyes.

Sample size is an important consideration in the development, testing and validation of a model based on machine learning. Previous studies that have applied machine learning to detect subclinical KC have typically presented with between 15 and 77 cases (median = 67) and control eyes ranging from 60 to 2980 (median = 177) (Table 2) [6,7,9–15]. The current study (145 eyes) had double the median number of subclinical KC eyes, and a comparable number of control eyes (122 eyes) compared to other studies. The study by Lopes et al. [14] presented with a very large number of control eyes (n = 2980) and reported a good specificity of 97%, that put it at the upper end of previous studies in being able to accurately detect controls. This is unsurprising, given the large proportion of negatives (control) in the collection. However, it achieved a low sensitivity of 80% which was at the lower end of previous studies, i.e. it could only correctly detect 80% of subclinical KC eyes. Overall, the performance indicated a bias towards negative predictions in the model, explaining both strong prediction of True Negatives, and more False Negatives. These results compared to the current study where we were able to obtain a much more balanced sensitivity of 97% and specificity of 98% providing an overall accuracy of 98%, over a more balanced dataset, with comparable numbers of cases and controls.

Table 2
Accuracy, sensitivity and specificity of built models in previous studies compared to the model built in the current study. NA indicated that the accuracy value was not provided in the study. Accuracy, sensitivity and specificity were obtained using the datasets in the columns, 'No. of case eyes' and 'No. of control eyes'. Bolded numbers indicated the result of the current study.

First Author	Accuracy	Sensitivity	Specificity	No. of case eyes	No. of control eyes
Saad et al. [6]	92%	93%	92%	40	72
Smadja et al. [7]	NA	94%	97%	47	177
Chan et al. [9]	93%	71%	98%	24	104
Kovacs et al. [10]	NA	90%	90%	15	60
Ruiz et al. [11]	93%	79%	98%	67	194
Ambrosio et al. [12]	93%	90%	96%	94	480
Xu et al. [13]	84%	84%	85%	77	147
Lopes et al. [14]	NA	80%	97%	71	2980
Issarti et al. [15]	97%	98%	96%	77	312
Current Study	**98%**	**97%**	**98%**	**145**	**122**

When compared to our research, the study conducted by Issarti et al. showed a slightly higher sensitivity value [15]. Their results were achieved via the use of a 10-fold cross validation procedure on a smaller data set of 77 early KC eyes, meaning that each fold tested had only 7–8 eyes. In our research, we included 145 subclinical KC eyes, and we utilized 10-fold cross validation in training data to choose the best model. Importantly, our final evaluation was performed on an independent test dataset of 81 eyes to determine the model's performance, a stronger test of generalization of the model. We are therefore able to report with more confidence on our current model's performance as a consequence of these aspects of our evaluation.

4.1. Application

One of the challenges of any machine learning method is to develop a proof-of-concept clinical decision support tool for incorporation into the clinic. Such a process could be described as presented in Supplementary figure 2. This was based on using the Pentacam parameter output for an assessed eye. The output parameters would then undergo processing by our program including data cleaning, dimensionality reduction, random forest model and ultimately, a diagnostic result would be displayed. This application could be incorporated as an adjunct measure in the detection of subclinical KC alongside other clinical measures collected by the clinician, e.g., vision and refraction and family history that could form a comprehensive diagnosis process for subclinical KC that would allow the clinician a more robust basis for future treatment decisions.

4.2. Limitations

The current study has several limitations. Firstly, our model did not include other clinical measures such as vision and refraction, nor demographic data such as sex, age and ethnic background, which may have an impact in the detection of subclinical KC. Additionally, the present research findings were derived from the use of Pentacam parameters. The exportability of developed models to other corneal topography machines requires additional examination of parameter consistency and reproducibility across various corneal topography systems and is a planned extension of the current study. Our experience in developing the current model through the analysis of a high volume of Pentacam parameters plus the analysis procedure we presented in the current study could be easily adapted for the inclusion of additional data sources.

Another limitation of the study is that our model has not been tested on an external dataset and thus overall performance could be impacted. Any impact on performance could arise from variations in study design, machine learning methods employed, different participant groups and potential differences in how clinicians might label eyes between studies. This is a common issue in the machine learning field. To ensure the robustness of our model, additional internal and external datasets are expected to be collected and analysed in future studies, with the aim to more rigorously test our model when presented with data from other clinics, clinicians and methodologies.

5. Conclusions

In the current study, we have gone some way to developing a new clinical decision-making system based on the use of machine learning and utilizing the complete Pentacam output parameters in the diagnosis of subclinical KC. The established model can detect subclinical KC in a fully automatic manner with high accuracy, specificity and sensitivity. We have provided very promising results in classifying control and subclinical KC eyes, achieving 98% accuracy on a held-out test set. The ultimate benefit of this system will be to assist in the clinical decision process in terms of diagnosis and treatment paradigms available for the patient with the goal of preserving as much vision as possible.

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Declaration of competing interest

None for any authors.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2021.104884.

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